

Return vs. violation rate (final checkpoint, 312 episodes/seed; hollow = soft cost > 25/episode)



● SafeOT-typed (ours) ▼ SDAC (multi-constraint SOTA) SAC-Pen coefficient sweep
 ■ FDPI (constraint-based) + SRCPO (multi-constraint SOTA) --- FDPI p_f sweep
 ◆ SAC-Lagrangian (penalty-based) — typed Pareto frontier (soft ≤ 25) --- SAC-Lag cost-limit sweep
 ▲ SAC + fixed penalty (penalty-based)